

3.7.5 Magnetic fields (A-level only)

3.7.5.1 Magnetic flux density (A-level only)

Content	Opportunities for skills development
Force on a current-carrying wire in a magnetic field: $F = BIl$ when field is perpendicular to current. Fleming's left hand rule. Magnetic flux density B and definition of the tesla.	
Required practical 10: Investigate how the force on a wire varies with flux density, current and length of wire using a top pan balance.	

3.7.5.2 Moving charges in a magnetic field (A-level only)

Content	Opportunities for skills development
Force on charged particles moving in a magnetic field, $F = BQv$ when the field is perpendicular to velocity. Direction of force on positive and negative charged particles. Circular path of particles; application in devices such as the cyclotron.	MS 4.3 Convert between 2D representations and 3D situations.

3.7.5.3 Magnetic flux and flux linkage (A-level only)

Content	Opportunities for skills development
Magnetic flux defined by $\Phi = BA$ where B is normal to A . Flux linkage as $N\Phi$ where N is the number of turns cutting the flux. Flux and flux linkage passing through a rectangular coil rotated in a magnetic field: flux linkage $N\Phi = BAN\cos\theta$	
Required practical 11: Investigate, using a search coil and oscilloscope, the effect on magnetic flux linkage of varying the angle between a search coil and magnetic field direction.	

3.7.5.4 Electromagnetic induction (A-level only)

Content	Opportunities for skills development
Simple experimental phenomena. Faraday's and Lenz's laws. Magnitude of induced emf = rate of change of flux linkage $\varepsilon = N \frac{\Delta \Phi}{\Delta t}$ Applications such as a straight conductor moving in a magnetic field. emf induced in a coil rotating uniformly in a magnetic field: $\varepsilon = BAN\omega \sin \omega t$	

3.7.5.5 Alternating currents (A-level only)

Content	Opportunities for skills development
Sinusoidal voltages and currents only; root mean square, peak and peak-to-peak values for sinusoidal waveforms only. $I_{\text{rms}} = \frac{I_0}{\sqrt{2}} ; V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$ Application to the calculation of mains electricity peak and peak-to-peak voltage values. Use of an oscilloscope as a dc and ac voltmeter, to measure time intervals and frequencies, and to display ac waveforms. No details of the structure of the instrument are required but familiarity with the operation of the controls is expected.	

3.7.5.6 The operation of a transformer (A-level only)

Content	Opportunities for skills development
The transformer equation: $\frac{N_s}{N_p} = \frac{V_s}{V_p}$ Transformer efficiency = $\frac{I_s V_s}{I_p V_p}$ Production of eddy currents. Causes of inefficiencies in a transformer. Transmission of electrical power at high voltage including calculations of power loss in transmission lines.	MS 0.3 / AT b, h Investigate relationships between currents, voltages and numbers of coils in transformers.